# Leo McDonnell

leomcd@berkeley.edu | Berkeley, CA | linkedin.com/in/leo-mcdonnell | github.com/leomcd

## Education

**University of California, Berkeley** — Expected May 2030
*Bachelor of Arts in Computer Science* — *Berkeley, CA*

## Technical Skills

**Languages**: Python, JavaScript, C++, Java, HTML, CSS
**Frameworks & Tools**: PyTorch, React, Flask, AWS, Git, MLflow, Docker
**Areas**: Machine Learning, Computer Vision, AI Interpretability, Full-Stack Development

## Work and Research Experience

**AI Research Intern** — Jul. 2026 – Present
*Stanford University – Sarin Lab*
- Trained a medical image regression model to identify anatomical landmarks on body parts.
- Incorporated this model into an app for deployment in clinical settings.
- Gathered feedback from app beta testers and contributed to product discussions.

**Summer Research Intern** — Jun. – Jul. 2025
*Crabel Capital Management, LLC*
- Developed Python research infrastructure to analyze historical market price movements.
- Presented research methodology, empirical findings, and potential applications and strategies to the firm's core research team at the conclusion of the internship.

**Paid Fellowship** — 2025 – 2026
*California Young Artists Symphony*
- Performed as a violinist in a pre-professional orchestra alongside professional musicians.

## Projects

**Interpreting LSTMs and Transformers on Latent-State Sequence Tasks** — Jun. 2026 – Present
- Conducted independent research, comparing two architectures using linear probes, representational similarity analysis, causal interventions, and visualization techniques.
- Investigated differences in internal representations, drafting a manuscript on interpretability.

**Tree Failure Analysis** — 2024 – 2025
- Collaborated with a consulting arborist to predict tree failure using machine learning.
- Performed data preprocessing, exploratory analysis, Random Forest modeling achieving over 90% accuracy, and model interpretation to support future tree risk assessments.

**MindForm AI** — 2025 – Present
- Developed a Visual Studio Code extension that presents an interactive, high-level graph automatically and bidirectionally synced with source code using the Google Gemini API.

**Handwritten Parser and C++ Transpiler** — 2025 – Present
- Built a source-to-source compiler project implementing a custom language grammar, handwritten parser, abstract syntax tree generation, semantic analysis, and C++ generation.

**Custom 2D/3D Game Engine**
- Developed an object-oriented, cross-platform game engine with a custom rendering pipeline, editor tooling, asset management, and physics integration.

## Publications

**Equal contributor, "Deep-Learning and Analytical Methods to Spatialize Stereoscopic Images"** 2025
- Published in the *National High School Journal of Science*. Developed a hybrid stereoscopic image architecture combining Pix2Pix, depth estimation, and interpolation, achieving an improvement in SSIM from 0.76 to 0.81 in the output image.

## Prizes and Awards

**Ventura County Hackathon by the Sea** – 3rd Place and Best in Environmental Sustainability for a PyTorch convolutional network to identify plant diseases — 2024

**Eagle Scout**, Scouting America — 2023